

Immifin Architecture Decision Record (ADR)

Version: 1.0

Purpose:

This document records important architectural decisions made during the development of Immifin.

Each decision includes:

- Context
- Decision
- Reasoning
- Trade-offs
- Future review

This document should evolve over time. Decisions are never deleted; if they change, add a new ADR explaining why.

ADR-001

Authentication Provider

Status

Accepted

Date

June 2026

Decision

Use Clerk for authentication.

Reason

Clerk provides:

- Excellent Next.js support
- Passwordless login
- Email verification
- Social login

- User management
- Organizations (future)
- Good developer experience

Alternatives Considered

- NextAuth/Auth.js
- Firebase Authentication
- Supabase Auth

Why Not Supabase Auth?

We wanted:

- Simpler frontend integration
- Better user management
- Easier production scaling

Supabase is used only as the application database.

Review

If Clerk pricing or limitations become unacceptable, evaluate Auth.js.

ADR-002

Database

Status

Accepted

Decision

Use Supabase PostgreSQL.

Reason

Needed:

- PostgreSQL
- Managed backups
- SQL migrations
- Good TypeScript support

Avoided self-hosting.

ADR-003

Hosting

Status

Accepted

Decision

Use Cloudflare Pages.

Reason

Advantages

- Low cost
- Fast global CDN
- Excellent caching
- Easy GitHub deployment

Alternatives

- Vercel
- Netlify
- AWS

Cloudflare best matched expected traffic and operating cost.

ADR-004

Visa Bulletin Source

Status

Accepted

Decision

Google Sheets is the source of truth.

Website always reads published CSV.

Reason

Advantages

- Easy manual editing
- Non-technical admin friendly
- Version history
- No direct database editing

Future

May migrate historical archive into PostgreSQL if editing or analytics become complex.

ADR-005

Data Refresh Strategy

Status

Accepted

Decision

Pull architecture.

Admin clicks:

Refresh Visa Bulletin

↓

Website downloads latest published CSV

↓

Database/cache updated

Not Used

Automatic push from Google Sheets.

Reason

Manual approval prevents accidental publication of incorrect data.

ADR-006

Public Data Caching

Status

Accepted

Decision

Use multiple cache layers.

Layers

Cloudflare CDN

↓

Next.js revalidate

↓

SWR in browser

Reason

Visa Bulletin changes monthly.

Aggressive caching greatly reduces infrastructure cost.

ADR-007

Client Data Fetching

Status

Accepted

Decision

Use SWR.

Alternatives

React Query

Apollo

Manual fetch

Reason

Simple API.

Perfect for immutable data.

Small bundle size.

ADR-008

Subscription Model

Status

Accepted

Plans

Guest

Free

Basic

Pro

Admin

Philosophy

Guest

Public information.

Free

Registered users.

Basic

Premium alerts.

Pro

AI assistant.

Admin

Internal management.

ADR-009

AI Strategy

Status

Accepted

Decision

LLM provider should be replaceable.

Primary

OpenAI

Future

Anthropic

Gemini

Other providers

Application code should depend on an internal AI abstraction layer rather than a vendor SDK.

ADR-010

Admin Portal

Status

Accepted

Decision

Admin operations happen inside Immifin.

Examples

Refresh Visa Bulletin

Manage users

Audit logs

Subscriptions

Reason

Avoid running scripts manually.

Everything should be accessible through the website.

ADR-011

Environment Strategy

Status

Accepted

Development

Clerk Development

Supabase Dev

Cloudflare Preview

Stripe Test

Production

Clerk Production

Supabase Production

Cloudflare Production

Stripe Live

Reason

Prevent accidental production changes during development.

ADR-012

Documentation First

Status

Accepted

Decision

Major architecture decisions must be documented before implementation.

Affected documents

Architecture

Database

Deployment

Roadmap

ADR

Reason

Documentation is treated as part of the product.

ADR-013

Build Philosophy

Status

Accepted

Guiding Principles

- Build slowly.
 - Prefer maintainability over cleverness.
 - Optimize for correctness before optimization.
 - Minimize monthly operating cost.
 - Avoid vendor lock-in where practical.
 - Keep the system understandable for a single developer.
 - Every feature should provide measurable value to immigrants.
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Future ADRs

Potential future decisions include:

ADR-014

RAG architecture

ADR-015

Vector database selection

ADR-016

WhatsApp integration

ADR-017

Push notification service

ADR-018

Payment provider expansion

ADR-019

Attorney portal

ADR-020

Multi-language support

How to Add a New ADR

Every new ADR should contain:

Title

Status

Date

Context

Decision

Alternatives

Reasoning

Trade-offs

Future Review

Never edit historical decisions.

If a decision changes, create a new ADR that supersedes the previous one and explain why.